

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-3: Implement dynamic web applications by utilizing DOM-based client-side scripting and

employing Java Servlets for server-side request processing and response handling. (L3)

Assessment Tool: Technical Presentation

Weightage: 20%

Code of Conduct:

I P.Sameera Banu(192372172) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sameera banu

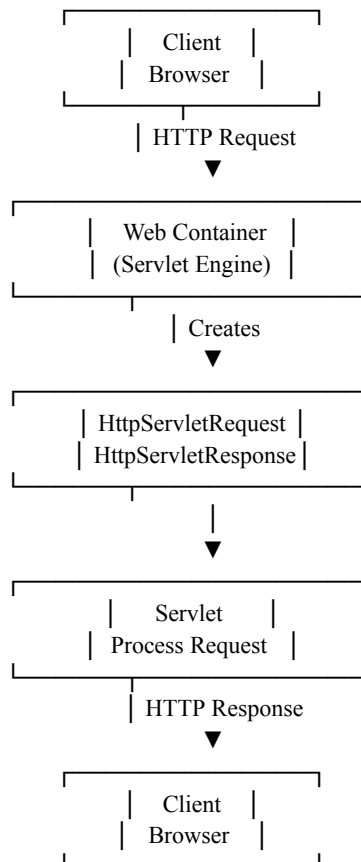
INTERNET PROGRAMMING

SERVLET TOPICS

1. Servlet Architecture & Workflow

Definition: A Servlet is a Java program that runs on the web server and handles client requests.

Architecture / Request–Response Cycle



Request–Response Steps

- User sends a request from the browser.
- The Web Container receives the HTTP request.
- The container creates HttpServletRequest and HttpServletResponse objects.
- The Servlet processes the request and may access a database.
- The Servlet generates the response.
- The Web Container sends the HTTP response back to the browser.

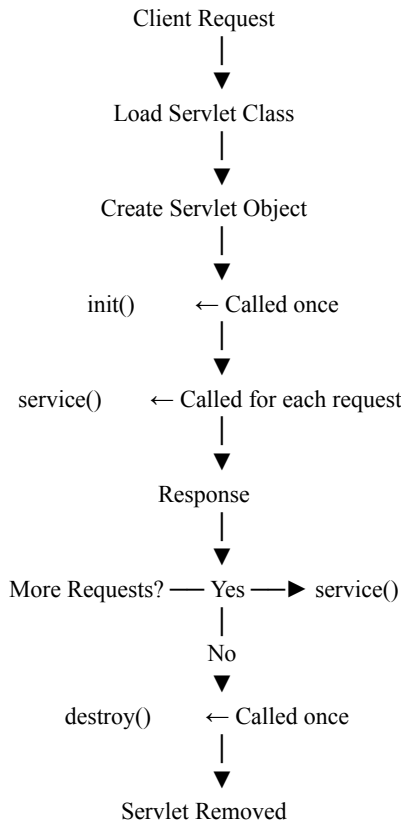
Role of Components

- Client: Sends HTTP requests and displays responses.

- Web Container: Manages servlet execution, life cycle, and request handling.
- Request Object: Contains form data, parameters, headers, and other request information.
- Response Object: Used to send output, status codes, headers, and content to the client.

2. Servlet Life Cycle Analysis

The Servlet Life Cycle describes how a servlet is created, used, and removed by the Web Container.



Method	Purpose	Execution
init()	Initializes the servlet and performs one-time setup.	Once
service()	Processes client requests and generates responses.	For every request
destroy()	Releases resources before the servlet is removed.	Once

Performance Implications

- init() should be used for one-time initialization so repeated setup is avoided.
- service() is executed frequently, so its code should be efficient.
- destroy() should close database connections, files, or other resources to avoid memory/resource leaks.

3. GET vs POST Method Evaluation

GET and POST are HTTP methods commonly used when submitting data from HTML forms.

Feature	GET	POST
Data transmission	Data is sent in the URL as query parameters.	Data is sent in the HTTP request body.
Visibility	Data is visible in the URL.	Data is not normally displayed in the URL.
Security	Not suitable for sensitive data.	Better for sensitive data, but HTTPS is still required.
URL limitation	Limited by browser/server URL length.	Can send much larger amounts of data.
Bookmarking	Can be bookmarked.	Normally not bookmarked as a reusable request.
Typical use	Search, filtering, viewing data.	Login, registration, uploading, form submission.

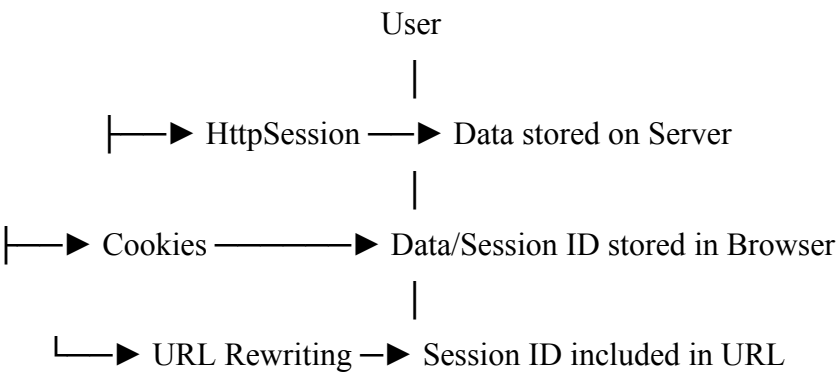
Appropriate Scenarios

- GET: Searching for a product, viewing a page, filtering results, or retrieving information.
- POST: Login, registration, submitting feedback, uploading data, or processing a transaction.

Important: POST is not automatically secure. Sensitive information should be transmitted over HTTPS.

4. Session Tracking Techniques

Session tracking is used to maintain information about a user across multiple HTTP requests.



Technique	Advantages	Limitations / Security
HttpSession	Easy to use; session data remains on server; good for logged-in users.	Consumes server memory; session IDs must be protected.
Cookies	Simple; can persist across visits; useful for preferences.	Can be disabled or deleted; vulnerable if sensitive data is stored directly.

URL Rewriting	Works when cookies are unavailable.	URLs become longer; session IDs in URLs can create security/privacy risks.
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Use-Case Comparison

- HttpSession: Online banking, shopping carts, and authenticated applications.
- Cookies: Remembering preferences or a user's 'Remember Me' choice.
- URL Rewriting: Applications that need session tracking when cookies are unavailable.

5. Cookies vs Sessions – Critical Analysis

Cookies and sessions are both used to maintain state in web applications, but they store information in different places.

Feature	Cookies	Sessions
Storage location	Client/browser.	Server.
Security	Generally less secure if sensitive data is stored directly.	Generally safer because application data is stored on the server.
Lifetime	Can be session-based or persistent with an expiry time.	Usually lasts until timeout, logout, or session invalidation.
Performance	Sent with relevant requests, adding network overhead.	Server stores data; usually only a session identifier is sent by the client.
Best use	Preferences, language, theme, remembering users.	Login state, shopping carts, temporary user data.

Real-Time Use Cases

- Cookies: A shopping website remembers the user's language or theme preference.
- Sessions: An online banking website maintains the authenticated user session after login.
- Shopping Cart: A session can maintain the cart while the user moves between pages.

Quick Revision

Topic	Remember This
Servlet Architecture	Client → Web Container → Servlet → Response
Servlet Life Cycle	init() → service() → destroy()
GET vs POST	GET for retrieving data; POST for submitting data
Session Tracking	HttpSession, Cookies, URL Rewriting
Cookies vs Sessions	Cookie = client side; Session = server side